

EXPERIMENT-13

13. | Implement your own ping program.

ping_program.py

```
import socket
import time

host = input("Enter host/IP address: ")

try:
    ip = socket.gethostbyname(host)
    print("Pinging", host, "[", ip, "]")

    for i in range(4):
        start = time.time()

        try:
            s = socket.create_connection((ip, 80), timeout=2)

        end = time.time()

        print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")

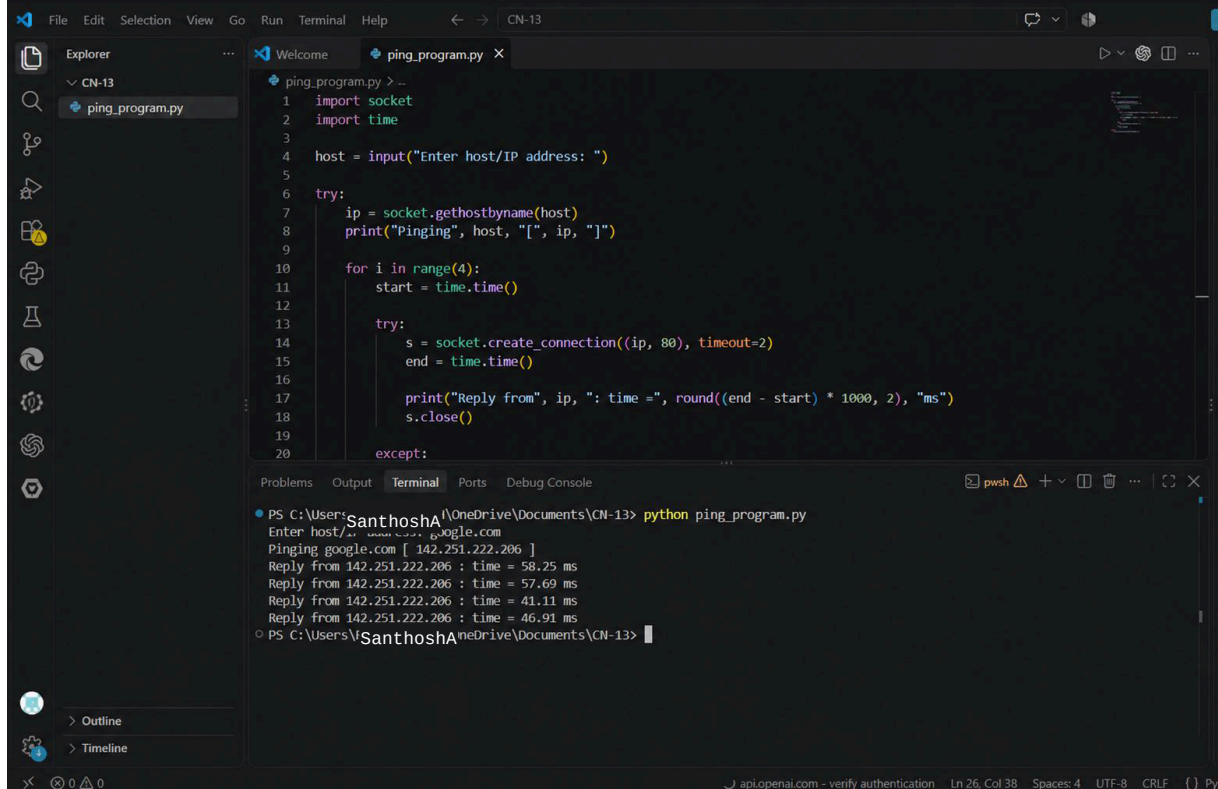
    s.close()

    except:
        print("Request timed out.")

        time.sleep(1)

except:
    print("Invalid host/IP address.")
```

```
PS C:\Users\SanthoshA\OneDrive\Documents\CN-13> python ping_program.py
Enter host/IP address: google.com
Pinging google.com [ 142.251.222.206 ]
Reply from 142.251.222.206 : time = 58.25 ms
Reply from 142.251.222.206 : time = 57.69 ms
Reply from 142.251.222.206 : time = 41.11 ms
Reply from 142.251.222.206 : time = 46.91 ms
PS C:\Users\SanthoshA\OneDrive\Documents\CN-13>
```



The screenshot displays the Visual Studio Code (VS Code) interface. The Explorer sidebar on the left shows the file structure with 'CN-13' and 'ping_program.py'. The main editor window shows the source code of 'ping_program.py' with the following content:

```
1 import socket
2 import time
3
4 host = input("Enter host/IP address: ")
5
6 try:
7     ip = socket.gethostbyname(host)
8     print("Pinging", host, "[", ip, "]")
9
10    for i in range(4):
11        start = time.time()
12
13        try:
14            s = socket.create_connection((ip, 80), timeout=2)
15            end = time.time()
16
17            print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")
18            s.close()
19
20    except:
```

The Terminal panel at the bottom shows the execution output, which matches the text shown in the first block of the image. The status bar at the bottom indicates the file is 'api.openai.com - verify authentication', line 26, column 38, with 4 spaces, UTF-8 encoding, and CRLF line endings.